ACTT: Airway

Capt Vince Soh, MD CCFP-EM
*Slides taken from LCdr Vuong (with minor changes)

Outline

- Review objectives and indications for airway management
- Describe airway assessment techniques
- Review options for airway intervention
- Hands-on practice

Objectives of Airway Management

- Ensure *ventilation* and *oxygenation*
  - Maintain airway *patency*
  - *Protect* airway against aspiration

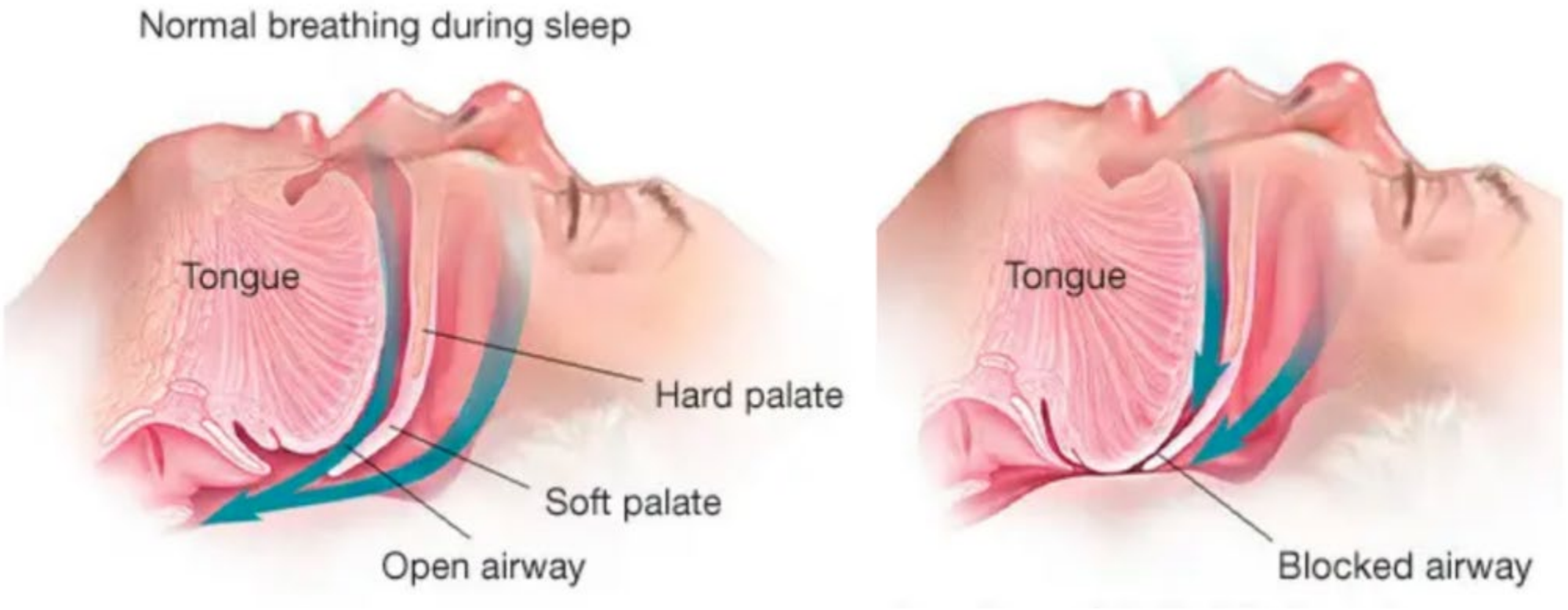
Indications for Airway Management

- Failure to OXYGENATE
- Failure to VENTILATE
- Failure to protect the airway
- Predicted clinical course

Examples…

- Obstructed airway
- Threatened airway
- Hypoxia or respiratory arrest
- Decreased level of consciousness (“GCS 8, intubate”)
- Facial/neck trauma

Indications for Airway Management

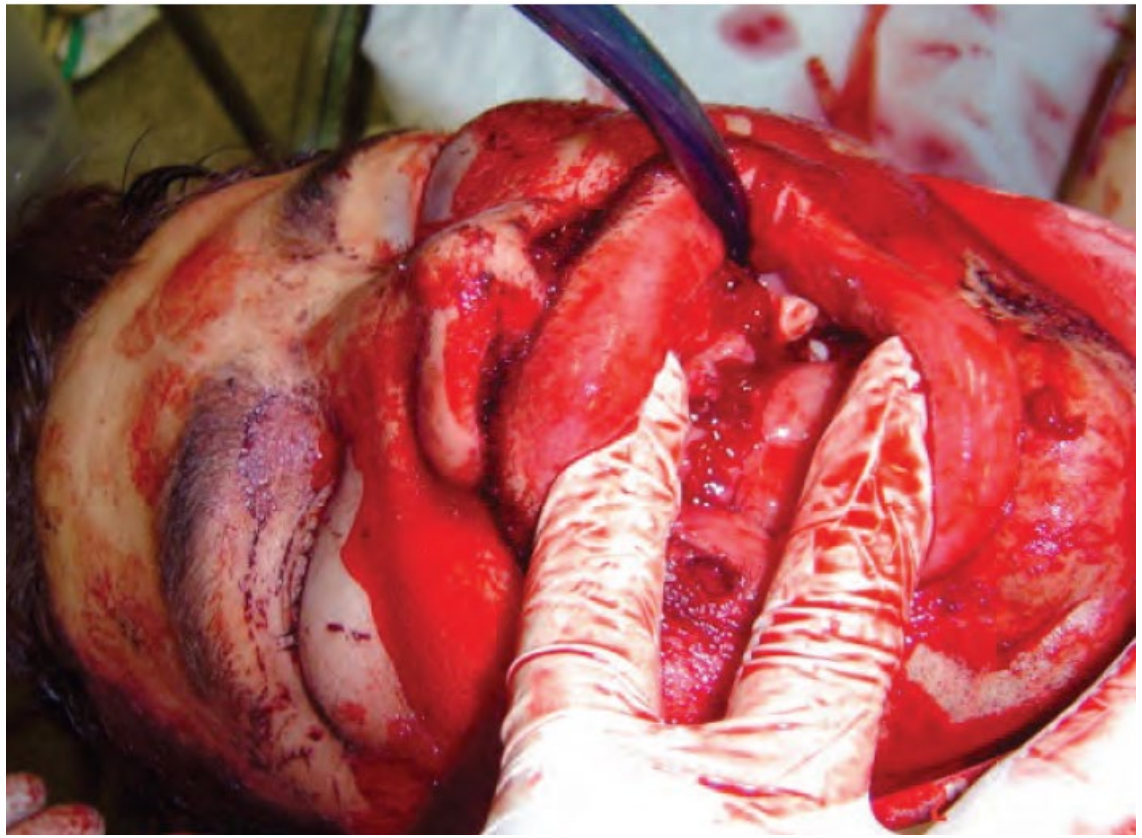


Mayoclinic.org

Indications for Airway Management

ATLS 10e

Indications for Airway Management



ATLS 10e

Deciding Intervention

- Do they need supplemental O_2?
- Do they need supportive measures?
- Do they need a supraglottic airway?
- Do they necessarily require a *definitive* airway?
  - A *definitive airway* is a tube placed in the trachea with the cuff inflated below the vocal cords, the tube connected to a form of oxygen-enriched assisted ventilation, and the airway secured in place with tape or other stabilizing method.

Planning, planning, planning…

- 7 Ps
- UK review of major airway-related complications/deaths found most common contributing factors:
  - Poor airway assessment
  - Poor planning
  - Lack of personal and/or institutional preparedness for managing difficult airways

Planning, planning, planning…

Summary: The Seven Ps of rapid sequence intubation

Action	Time
Preparation	10 minutes before intubation
Preoxygenation	5 minutes before intubation
Physiologic optimization	3 minutes before intubation (may be longer depending on necessary interventions and time available)
Paralysis with induction	Induction
Positioning with protection	30 seconds after induction
Placement (intubation) with proof	45 seconds after induction
Post-intubation management	60 seconds after induction

Preparation

- Primary airway plan
  - Experience of provider, backup providers
  - Location
  - Equipment
  - Airway assessment
  - Physiologic factors
- Difficult airway plan(s)
- Post intubation plan
  - Location
  - Equipment
  - Sedation
- Assign roles
- Use a checklist (e.g. “SOAPME”)
  - **S** → Suction
  - **O** → Oxygen
  - **A** → Airway equipment
  - **P** → Pharmacology
  - **M** → Monitors
  - **E** → Escape plan

"Airway Time-Out" Checklist

Emergency Airway Checklist

	CODE Status?	Review MOST, obtain informed consent from patient/family
	Don PPE and assign Team Roles	N95, Eye protection, Gown, Gloves

Assess patient for difficult airway		
	BOOTS for difficult BMV	**B**eard - **O**ld - **O**SA - no **T**eeth - **S**tridor
	LEMON for difficult intubation	**L**ook externally - Evaluate 3-3-2 - **M**allampati - **O**bstruction - **N**eck movement

Optimally Position Patient		
	Auditory meatus in line with sternal notch	face parallel to trunk/ceiling
	Elevate head + torso by 30 degrees	To help ventilate and reduce risk of aspiration
	Ramp up obese patient	Using multiple pillows and blankets, line up auditory meatus to sternal notch
	Dentures	**IN** for BVM, **OUT** for intubation

Pre-Oxygenate		
	Denitrogenation	**8** deep breaths OR **3 min.** of regular breathing on NR mask + NP (@15 L/min)
	Oxygenation	aim for minimum of **95%** O2 Sat.; keep NP on until intubated
	Sedate with Ketamine (DSI) if patient too combative to optimize pre-oxygenation	

Resuscitate before you intubate!		
	Hypotension (both treat and prevent)	See reverse for details
	Hypoxia	
	Metabolic acidosis	
	Increased ICP	
	Asthma/COPD	

Equipment = "MIDSOLES" (for pediatric patient, use Broselow Tape or Pedi STAT App)	
	Monitors (cardiac, pulse Ox, BP q5min)
	IV x 2, ?IO
	Drugs (See below)
	Suction x 2, under head of bed
	Oxygen (NR/NP at 15 L/min, BVM with attached PEEP valve and Capnometry, appropriately-sized **KING** Airway)
	Laryngoscope with appropriate blade (usually Mac 3, Mac 4 if large patient); King Vision video-laryngoscope - test light/power
	ET tubes, 8.0 (best for ICU) and 7.5, with 10mL syringe and lube for cuff; test balloon for leaks
	Stylet (loaded in ET tube of choice), **S**ecuring device for ET tube (or tape)
	Surgical airway equipment (#10 scalpel, Bougie, 6.0 ET tube or tube from Cric set)

Medications		
	Induction	**Ketamine** (10mg/mL) 1.5mg/Kg -> **100mg** for 70Kg adult (best if hypotensive)
		OR **Propofol** (10mg/mL) 1.5mg/Kg -> **100mg** for 70Kg adult (use less if hypotensive)
		OR Mix Ketamine + Propofol (**Ketofol**)
	Paralysis	**Rocuronium** (10mg/mL) 1mg/Kg -> **70mg** for 70Kg adult (use more if hypotensive)
	Push-Dose Pressors	**Epinephrine** (1mL of 1:10,000 Epi + 9mL NS = 10mL of Epi 10mcg/mL) **1mL q2-5min**
		OR **Phenylephrine** 50mcg/mL (premixed) **1-2mL q2-5min**
	Post-Intubation Sedation	**Ketamine** bolus 1mg/Kg then 1mg/Kg/hr
		OR **Propofol** bolus 1mg/Kg then 20-40 mcg/Kg/min
		OR **Morphine** (0.07mg/Kg) + **Midazolam** (0.1mg/Kg) -> 5mg/hr + 5mg/hr for 70Kg adult
	Ongoing pressure support	**Norepinephrine** 0.1mcg/Kg/min, titrate by 1mcg/min q5min

Determine best airway management strategy = PLAN A		
	Non-Invasive Positive Pressure Ventilation (NIPPV)	
	Awake Intubation	
	Delayed Sequence Intubation (DSI)	*Ketamine 0.5-1mg/Kg, + 0.5mg/Kg prn*
	Rapid Sequence Intubation (RSI)	
	Surgical Airway	

Review the PLAN with designated Assistant and Team		
	2-person BVM technique	Augment with **1-2 x NP** and **1 x OP** airways
	Guided BURP	**B**ackwards - **U**pwards - **R**ightwards - **P**ressure, guided by intubator's hand
	Guided head elevation	Can maintain position by folding hand towels under occiput
	Plan **B**	Bougie? switch blade? paralyse (if not done)? video-laryngoscope or DL?
	Plan **C**	Switch providers, verbalize possibility of a surgical airway
	Plan **D** -> King Airway or Surgical Airway	Mark cricothyroid membrane, have equipment ready, assign roles

emergencybccare.ca

Airway Assessment

- During initial airway assessment, a “talking patient” with appropriate verbal response provides *momentary* reassurance
  - Airway is patent
  - Ventilation intact
  - Brain perfusion sufficient

Airway Assessment

Key History:

- *Hx of difficult airway*

- Aspiration risk
  - Last po intake (fasted adults: 6 hrs solids, 2 hrs clear fluids)
  - GERD, hiatial hernia
  - Delayed gastric emptying → gastroparesis, medications (GLP-1 agonists), Hx bariatric surgery

- Medical conditions
  - Rheumatoid arthritis → decreased mouth opening, neck stiffness, unstable C-spine
  - Ankylosing spondylitis → neck stiffness, TMJ stiffness

- Prior head/neck radiation, previous trauma, known masses

Airway Assessment

Predictors of Difficult Bag-Mask Ventilation (BVM)

BOOTS mnemonic:

- **B** → Beard
- **O** → Obese
- **O** → Older patient
- **T** → Toothless
- **S** → Snoring/stridor

Predictors of Difficult Intubation

LEMON mnemonic:

- **L** → Look externally
- **E** → Evaluate 3-3-2
- **M** → Mallampati
- **O** → Obstruction
- **N** → Neck mobility

Airway Assessment

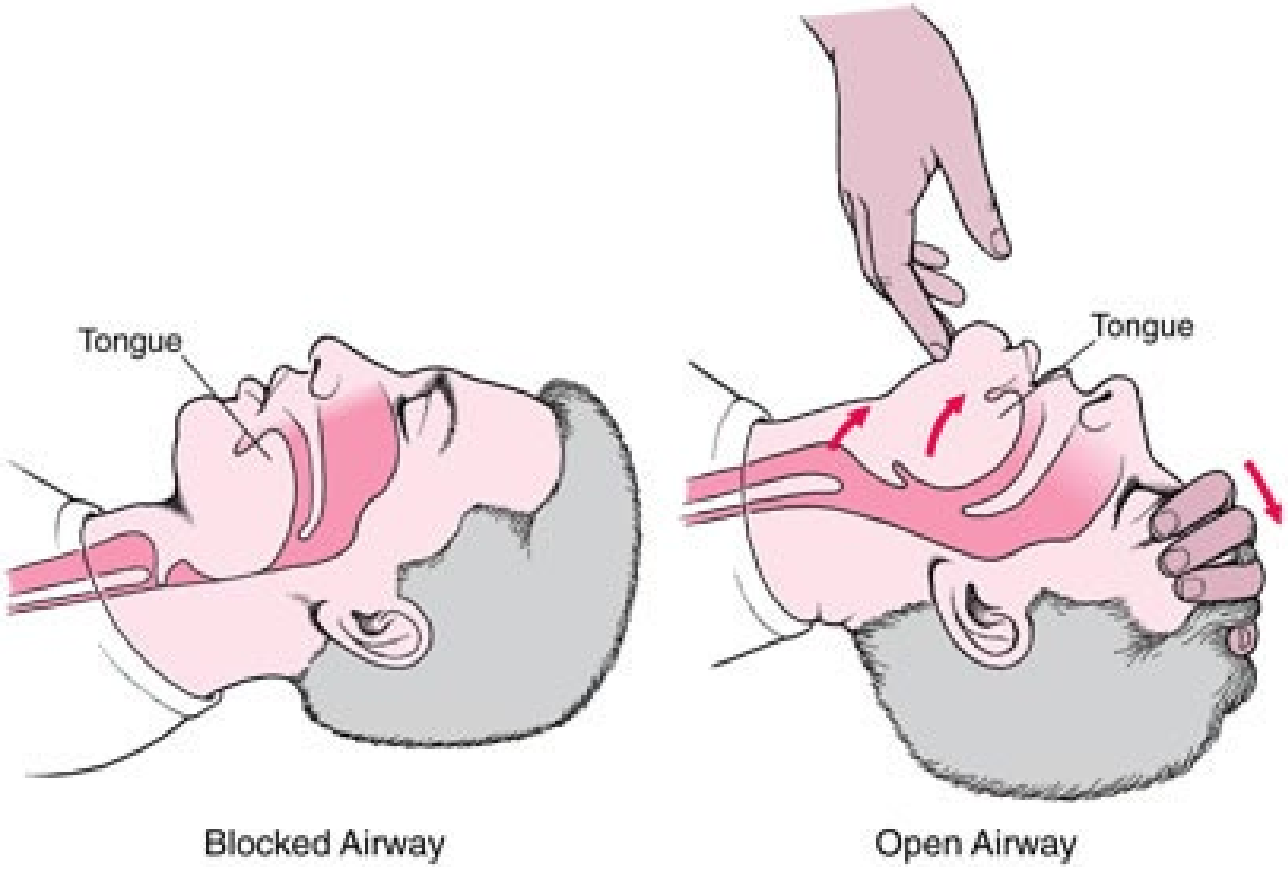
- Abbreviated depending on patient condition (e.g. trauma, impending airway compromise), environment (e.g. mass casualty)

Interventions

- Airway maneuvers
- Supplemental oxygen
- Bag-valve mask assisted ventilation with oral/nasal airway
- Supraglottic airway
- Endotracheal intubation
- Front-of-the-neck access (FONA) cricothyrotomy

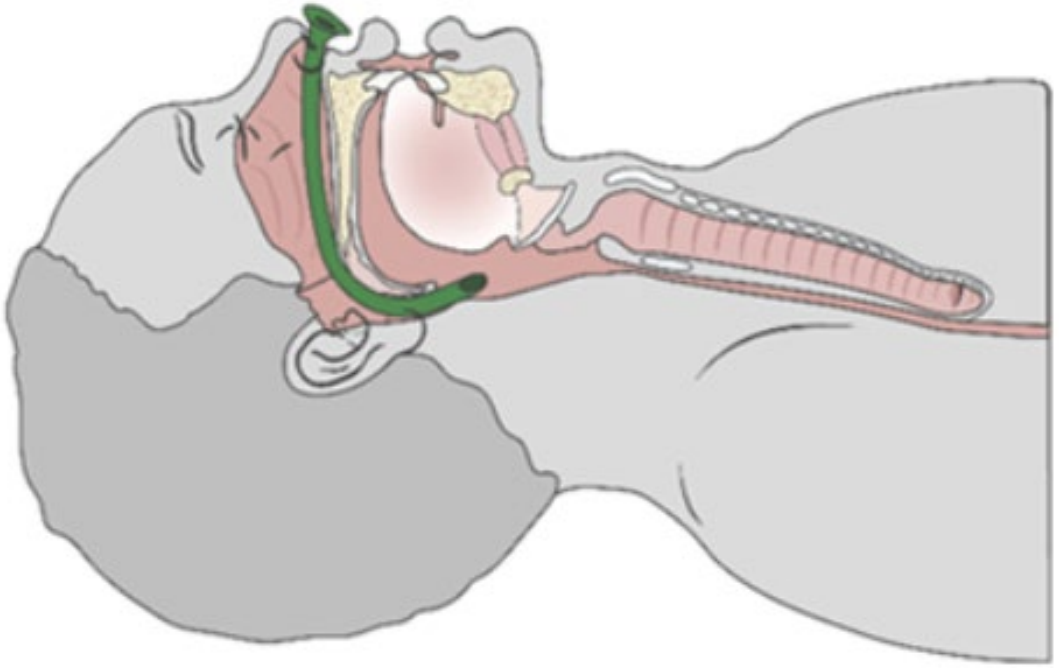
Open the Airway

- Head extension
- Chin lift
- Jaw thrust



Maintain Airway Patency

- Positioning
  - Ramping
  - Sniffing position
  - Head extension
- Adjuncts
  - Oral airway → no airway reflexes
  - Nasal airway → intact airway reflexes



Supplemental Oxygen

NONINVASIVE OXYGEN DELIVERY SYSTEMS

Device: Nasal Cannula
Flow: 1 - 6 L/min
FiO2: 25 - 40%
(~4%/L of flow)

Device: Face Mask
Flow: 5 - 10 L/min
FiO2: 40 - 60%

Device: Face Tent
Flow: 10 - 15 L/min
FiO2: ~40%

Device: Venturi Mask
Flow: 2 - 15 L/min
(based on valve)
FiO2: 24 - 60%
(precisely controlled)

Device: Non-Rebreather
Flow: 12 - 15 L/min
FiO2: 80 - 95%

Device: High Flow Nasal Cannula
Flow: up to 60 L/min
FiO2: 21 - 95%

Bag-Mask Ventilation

Supraglottic Airways

- Devices used to maintain airway patency *above level of cords* (i.e. supraglottic) without entering trachea

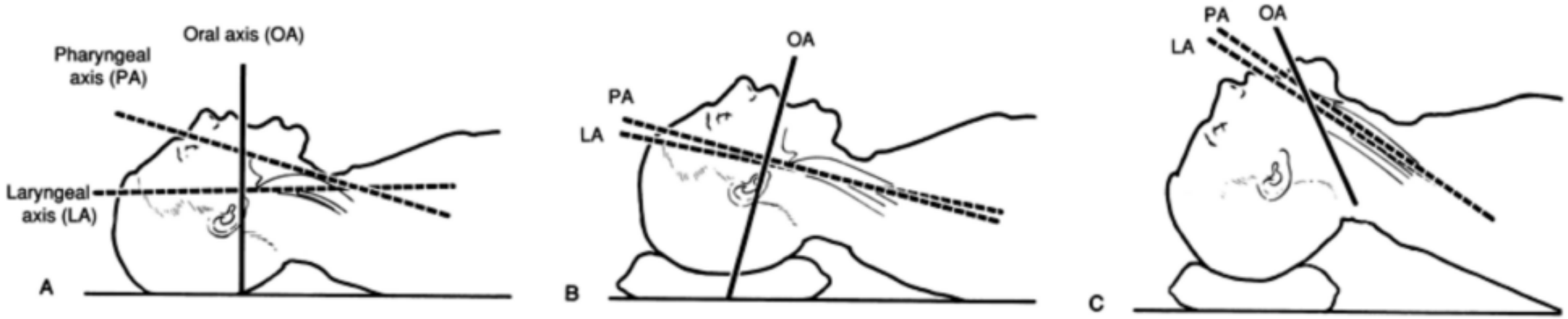
- Various types, including:
  - Laryngeal mask airway (LMA)
  - iGel
  - King LT
  - Combitube

- Can be rapidly inserted, but…
  - Does not protect against aspiration
  - Can dislodge/leak (not suited for long-term use)
  - Not ideal for high airway pressures, e.g. asthma

<u>Do not forget about supraglottic airways as a rescue if encountering a difficult airway!</u>

Endotracheal Intubation

- Cannot emphasize positioning enough for success



Endotracheal Intubation

TABLE 2-1 INDICATIONS FOR DEFINITIVE AIRWAY

NEED FOR AIRWAY PROTECTION	NEED FOR VENTILATION OR OXYGENATION
Severe maxillofacial fractures • Risk for aspiration from bleeding and/or vomiting	**Inadequate respiratory efforts** • Tachypnea • Hypoxia • Hypercarbia • Cyanosis • Combativeness
Neck injury • Neck hematoma • Laryngeal or tracheal injury • Inhalation injury from burns and facial burns • Stridor • Voice change	• Progressive change • Accessory muscle use • Respiratory muscle paralysis • Abdominal breathing
Head injury • Unconscious • Combative	• Acute neurological deterioration or herniation • Apnea from loss of consciousness or neuromuscular paralysis

ATLS 10e

Front-of-Neck Access (FONA)

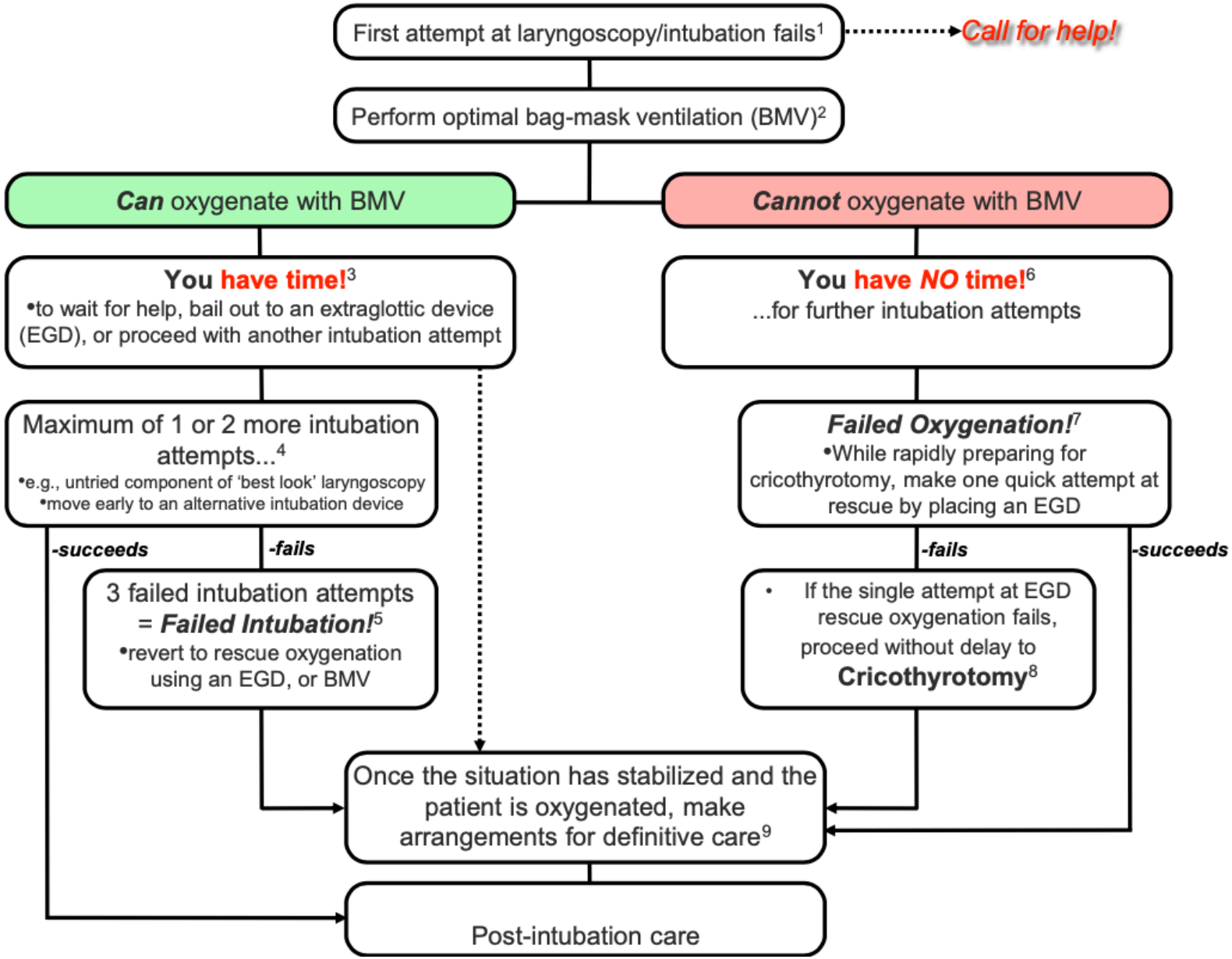
- CVCO: can’t ventilate, can’t oxygenate
- Age ≥8 or signs of puberty

10 Blade Scalpel

Bougie

6.0 ETT

…and a 10 mL syringe


First attempt at laryngoscopy/intubation fails[1]
Call for help!
Perform optimal bag-mask ventilation (BMV)[2]
Can oxygenate with BMV
Cannot oxygenate with BMV
You have time![3]
•to wait for help, bail out to an extraglottic device (EGD), or proceed with another intubation attempt
You have NO time![6]
...for further intubation attempts
Maximum of 1 or 2 more intubation attempts...[4]
•e.g., untried component of 'best look' laryngoscopy
•move early to an alternative intubation device
Failed Oxygenation![7]
•While rapidly preparing for cricothyrotomy, make one quick attempt at rescue by placing an EGD
-succeeds
-fails
-fails
-succeeds
3 failed intubation attempts = Failed Intubation![5]
•revert to rescue oxygenation using an EGD, or BMV
• If the single attempt at EGD rescue oxygenation fails, proceed without delay to Cricothyrotomy[8]
Once the situation has stabilized and the patient is oxygenated, make arrangements for definitive care[9]
Post-intubation care

Intubation Pearls

- Trauma patients challenging → C-spine collar limits airway assessment and manipulation
  - Manual in-line stabilization
- Optimize *positioning* (including yourself) – often forgotten about during traumas
- *Pre-oxygenate* to buy you more time during apneic period
  - Keep NP during intubation for apneic oxygenation
- Have *suction on and ready*
- Have *backups* and an *exit plan*

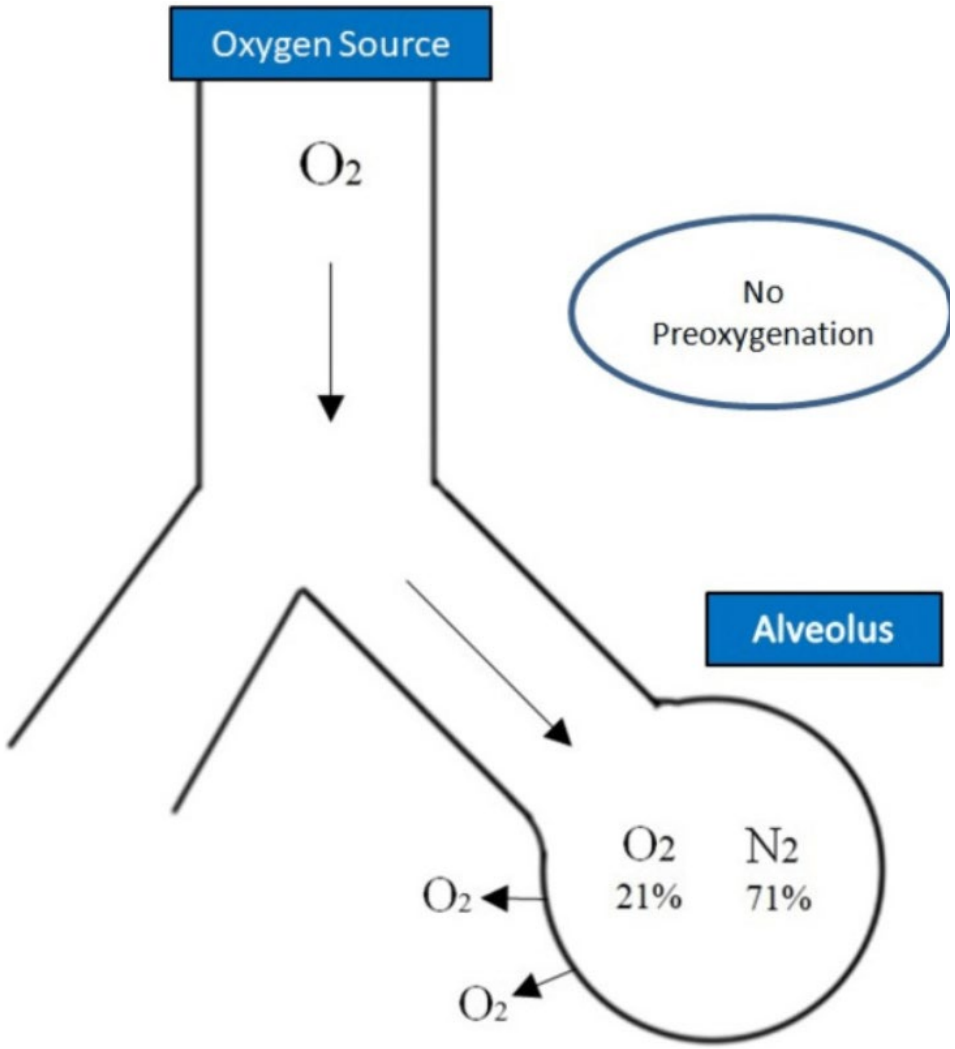
Preoxygenation

Goal is to denitrogenate alveoli. How to achieve this…

- If adequate respiratory drive:
  - Supplemental O2 via Face mask + NP (or BVM with PEEP)
  - >3-5 mins of tidal volume breathing, or
  - 8 full inspiration/expiration breaths
- If inadequate respiratory drive:
  - Support with BVM + supplemental O_2

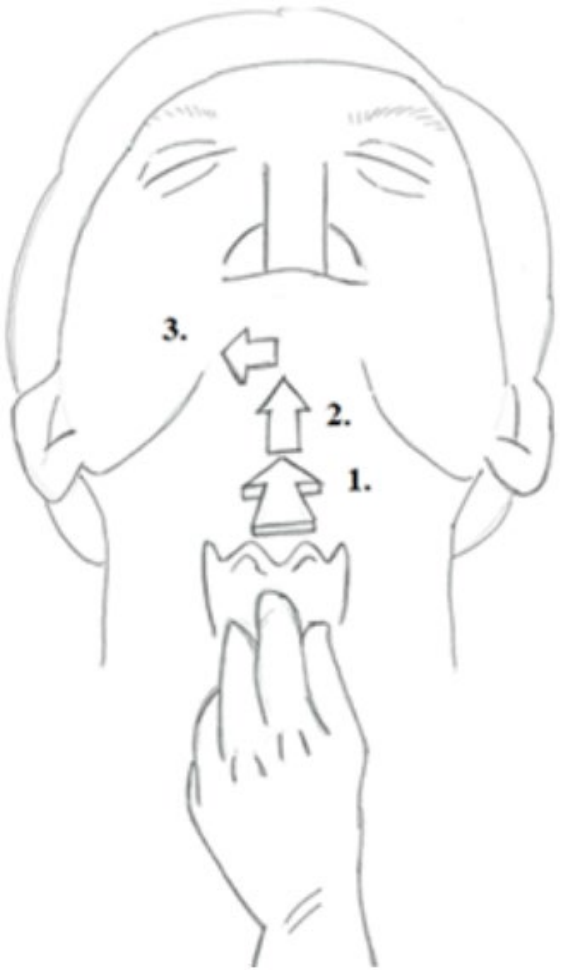
Apneic Oxygenation

- Principle of “apneic diffusion of oxygenation”
  - During apnea, oxygen continues to diffuse from alveoli into blood
  - Diffusion generates pressure differential between atmosphere and alveoli
  - Pressure gradient draws additional oxygen from administered oxygen source (i.e. NP) to alveoli



BURP (Backwards Upwards Rightwards Pressure)

- Maneuver to improve view on laryngoscopy
- Requires assistant → can guide assistant's hand during laryngoscopy
- Involves…
  - Displacing thyroid cartilage posteriorly until mild resistance
  - Superior by 2cm
  - Rightwards by 1-2cm



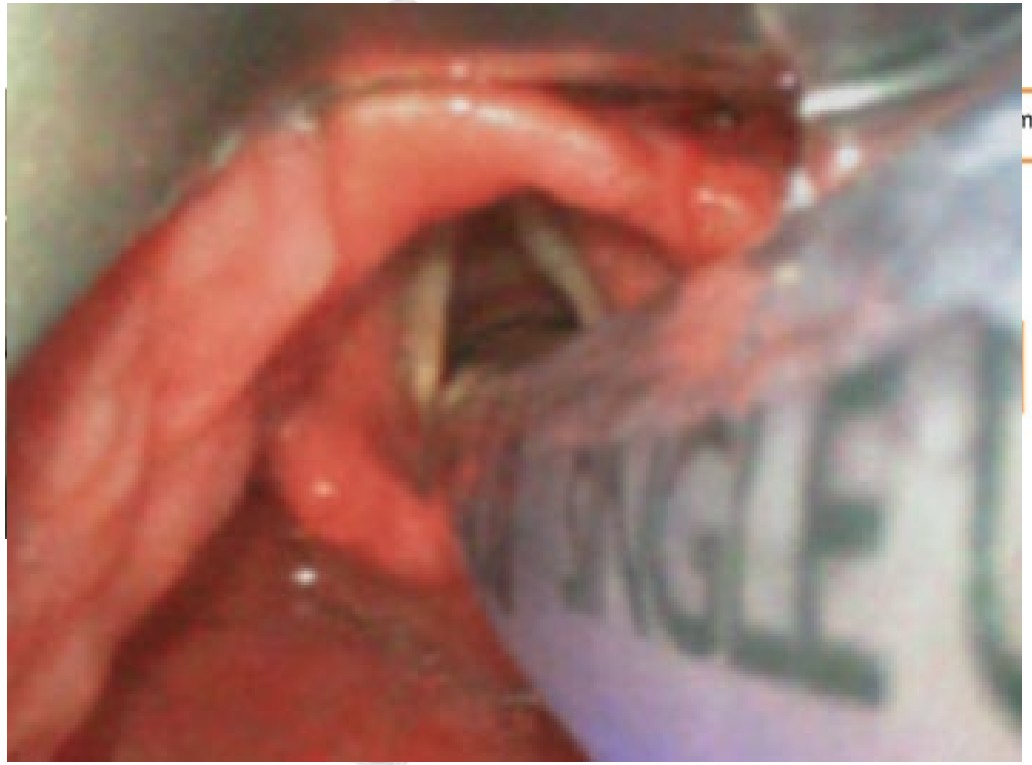
1. Backward
2. Upward
3. Rightward

ETT Placement Confirmation

Intubating the esophagus and not confirming placement will quickly kill your patient.

<u>Methods to confirm</u>:

- Waveform capnography → preferred
- Colorimetric CO_2 detectors
  - Yellow = YES
  - Purple = the colour of your pt when the tube is in the wrong place
- Visualizing ETT pass through cords



Immediate Post-Intubation Management

- Ventilator
- Support hemodynamics prn
- Keep patient sedated (especially if paralyzed)

Airway Management Competency

- Get hands-on practice as much as possible
- In controlled settings, 20-50 intubation attempts suggested to develop *competence*
  - Other literature suggests up to 250 intubations required for emergency physicians to become *proficient* in cardiac arrest patients (Carlson JN, 2019)
- Not just about intubation → ensure competency with airway positioning, oral/nasal airway insertion, BVM (1-person, 2-person techniques), LMA placement, cricothyrotomy
- On that note…

Questions?